

¹ Preschooler's systematic errors reveal conceptual knowledge and foreshadow word learning
² one year later

Abstract

Language learning involves the acquisition of word-referent mappings but also the organization of concepts into rich semantic networks. The study presented here uses children's systematic semantic errors during a language comprehension task as a window into their emerging networks. When making an error, picking a semantically related option signals some knowledge of the conceptual domain a word refers to. In a large, cross-sectional sample of preschool children ($N = 1103$), semantic errors became more frequent with age compared to other types of errors. Children who made more semantic errors also had larger vocabularies, suggesting that they signal an emerging conceptual knowledge that supports learning. This hypothesis was confirmed by showing that in a longitudinal sample ($N = 189$), making a semantic error predicted having learned a word one year later. Taken together, this research suggests that systematic semantic errors can be used as an early indicator of learning potential.

Keywords: language learning, errors, semantic networks

Word count: X

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Research Transparency Statement

General Disclosures

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Study Disclosures

Preregistration: The analysis plan was not pre-registered. Materials: The task used for data collection can be viewed via the following website: (masked). Data: All primary data are publicly available [*masked for peer review*]. Analysis scripts: All analysis scripts are publicly available [*masked for peer review*].

Introduction

Language and thought are tightly interconnected. While there ongoing fierce debates about whether language is necessary for thought and the extent to which learning and using a particular language shapes cognition (Fedorenko, Piantadosi, & Gibson, 2024; Lupyan, 2016; Majid, Bowerman, Kita, Haun, & Levinson, 2004), there is broad agreement that conceptual organization can be accessed through language. Semantic similarities and associations between words – semantic networks – closely mirror similarities and associations between concepts (Arunachalam & Waxman, 2010; Li, Zhao, & Mac Whinney, 2007; Lupyan & Lewis, 2019; Sloutsky & Deng, 2019; Unger & Fisher, 2021; Wojcik, 2018). The role of

language for the development of conceptual structure has been well documented (Sloutsky & Deng, 2019). The flip-side of this relationship – how conceptual structure supports language learning – has received considerably less attention and so have the potential cascading and self-reinforcing effects where learning new words supports conceptual development which in turn supports the learning of new words. Such effects have been found in other areas of language acquisition (Smith, Jones, Landau, Gershkoff-Stowe, & Samuelson, 2002), for example mutual exclusivity (Bohn, Tessler, Merrick, & Frank, 2021; Lewis, Cristiano, Lake, Kwan, & Frank, 2020) or syntactic bootstrapping (Babineau et al., 2024).

Research on semantic networks suggests that their structure shapes the order in which words are acquired. On this view, words form variable connections with other words, and the strength and structure of these connections determines which words are learned (Hills, Maouene, Maouene, Sheya, & Smith, 2009). For example, Fourtassi and colleagues (Fourtassi, Bian, & Frank, 2020) used data from ten languages to show that words that are more strongly connected to already known words are more likely to be acquired next. Yet, this line of research has largely characterized lexical structure at the level of words and populations rather than individual learners. Network structures are typically derived from adult data – for example, similarity ratings or free associations (McRae, Cree, Seidenberg, & McNorgan, 2005; Nelson, McEvoy, & Schreiber, 2004) – and aggregated across individuals. Whether these aggregate networks accurately reflect children’s developing semantic representations, and how such representations support language learning within individual children, is not well understood. The present study addresses this gap.

One – perhaps counter-intuitive – way to study semantic networks at the level of the individual is through children’s errors. In picture naming and comprehension tasks, children often select or produce semantically related words rather than unrelated alternatives (Altman, Goldstein, & Armon-Lotem, 2017). These semantic errors are systematic rather than random: they occur in both mono- and bilingual children (Krysztofiak et al., 2025) and

become increasingly common across the preschool years (Masterson, Druks, & Gallienne, 2008). Converging evidence comes from studies of semantic priming, which show that hearing a word activates related concepts already in infancy (Arias-Trejo & Plunkett, 2009). Furthermore, toddlers attend to semantically related referents even when the target word is not yet fully known (Wojcik, 2018). Taken together, these findings suggest that children may possess conceptual knowledge about a semantic domain before establishing precise word–referent mappings and that errors can be seen as systematic behavioral traces of such partial knowledge.

What remains unclear is whether the partial knowledge indexed by semantic errors provides a foundation for subsequent word learning. If lexical acquisition proceeds from broad conceptual knowledge to increasingly precise word meanings, then semantic errors should predict future acquisition. A child who selects a semantically related distractor may not yet know the target word, but may already represent the broader conceptual domain to which it belongs. The present study tested this idea using a combination of large cross-sectional and longitudinal datasets from preschool children completing a receptive vocabulary task. On each trial, children heard a noun or verb and selected one of several pictures, including a semantic distractor. This design allowed for examining both how semantic errors change across development and whether they predict subsequent word learning. The first analysis showed that semantic errors became more frequent with development. The second analysis found that semantic errors predicted overall vocabulary cross-sectionally, and the third analysis showed that making a semantic error predicted word learning one year later.

Methods

Participants

Participants were mostly monolingual German-speaking children living in two cities in the east (630k inhabitants) and north (75k inhabitants) of Germany. Parents had previously

indicated that they were interested in participating in studies on child development. The cross-sectional data included 1103 children (565 female, $m_{age} = 5.53$; range = 3.01 to 7.99). Data was either collected online ($N = 998$) or in person ($N = 105$). The longitudinal data set included 189 children (91 female, $m_{age} = 4.94$; range = 3.03 to 6.99), of which 176 were tested online and 13 in person. The data collected online has been reported in a previous study with a different focus (Bohn et al., 2024).

Material

All children completed the oREV (Bohn et al., 2024), an open-source receptive vocabulary task for German. The oREV was programmed as an interactive website, allowing for both online and in-person data collection. The original oREV included only nouns but has since been extended to also include verbs. The extended oREV included 22 nouns and 20 verbs. On each trial, children saw four drawings and heard a verbal prompt (“Zeige mir X”, eng.: show me X) asking them to select one of the drawings. The four drawings included the target picture, a semantic distractor, and two additional distractors. The oREV was originally designed to have a semantic, a phonetic, and an unrelated distractor. However, the phonetic distractor turned out to be difficult to realize because a) one cannot control if the child conceptualizes the drawing with the word that sounds familiar (the distractor showed a “Rohr”, eng.: pipe, but this could also be seen as a “Leitung”, eng.: line) and b) it was difficult to find a word that was phonologically similar and also was known to the child. As a consequence, many items ended up not having a phonological distractor, and we do not differentiate between types beyond the semantic distractor.

Procedure

For online data collection, families received a personalized link to complete the study at home. Parents were instructed to use a tablet or laptop (not a phone) and to guide the child through the study but not to interfere with the child’s responses. Children responded by either touching or using a mouse to click on a drawing. Responses were recorded and

coded automatically. Webcam recordings were collected to verify that the participant was indeed a child and to assess parental interference if necessary. Parents provided consent at the beginning of the study.

For in-person data collection, children were tested in a quiet room in their daycare by an experimenter using a tablet computer. The experimenter guided the child through the study but did not interfere with their choice. Children responded by touching a drawing on the touchscreen. Parents provided consent prior to the study.

Data analysis

The data were analyzed using Bayesian generalized linear mixed models fit in R using the package `brms` (Bürkner, 2017). All models used default priors. The first model predicted the probability of observing a semantic error (modeled using a Bernoulli distribution) by age (scaled to mean 0 and SD of 1) and word class with random intercepts for subject id and item and a random slope for age within item (model notation: `sem_dist ~ age + word_class + (1|id) + (age|item)`). The second model predicted the sum of correct trials (modeled as a binomial distribution) by age and the proportion of semantic errors (both scaled, model notation: `sum | trials(n) ~ age * prop_sem_errors`). The third, longitudinal, model predicted the probability of choosing the correct word (Bernoulli distribution) by age (scaled) and the distractor type chosen one year earlier (semantic or other) with random intercepts for id and item and random slopes for distractor type within id and age within item (model notation: `correct ~ age * distractor_type + (distractor_type|id) + (age|item)`).

Results

The first analysis showed that the proportion of semantic – compared to other – errors increased with age (estimate for age in a Bayesian GLMM: $\beta = 0.30$; 95% Credible Interval (CRI) [0.10 - 0.50]) and became dominant around 4.5 years of age (see Fig. 1A). This effect was robust across word classes, with no difference between nouns and verbs (estimate for

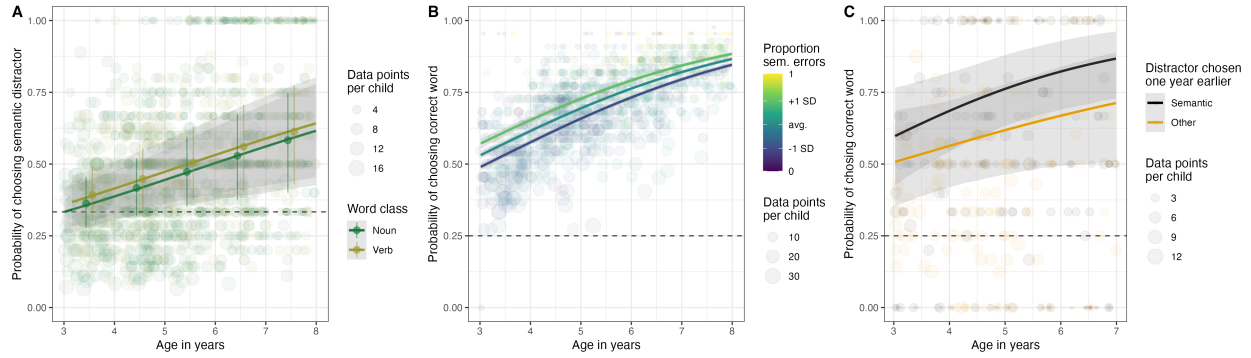


Figure 1. Cross-sectional and longitudinal links between semantic errors and vocabulary. A) Developmental trajectory for the probability to make a semantic error. Color shows word class (noun or verb). Lines show continuous model predictions and points with error bars show model predictions for each age bin of one year. B) Predicted probability of choosing a correct word in the task by the proportion of semantic errors made. Lines show model predictions for different levels of the continuous predictor. C) Probability of choosing a correct word conditional on the type of error made one year earlier. Lines show model predictions. Color shows error type (semantic or other). In all panels: error bands and bars show 95% credible intervals, transparent dots show aggregated data for each child with size proportional to the number of trials.

146 verbs: $\beta = 0.12$; 95% CRI [-0.42 - 0.65]).

147 In the second analysis, when predicting task performance by error type, we found
 148 children with a higher proportion of semantic errors to be more likely to choose a correct
 149 word (estimate for proportion of semantic errors: $\beta = 0.16$; 95% CRI [0.13 - 0.19]). This
 150 effect was strong, even when controlling for age and did not change with age (interaction
 151 between age and proportion of semantic errors: $\beta = 0.00$; 95% CRI [-0.03 - 0.03], see Fig.
 152 1B).

153 For the final analysis, we relied on the longitudinal dataset described above. The
 154 results showed that children were more likely to have learned a word if they had selected the

semantic distractor for this very word one year earlier (estimate for having picked semantic distractor: $\beta = 0.64$; 95% CRI [0.32 - 0.97]), This predictive relationship was robust across the preschool years (interaction btw. picking semantic distractor and age: $\beta = 0.18$; 95% CRI [-0.14 - 0.51]) Fig. 1C).

Discussion

The present study provides evidence that children’s errors in a receptive vocabulary task become increasingly systematic. With age, children favored semantically related distractors over unrelated alternatives, replicating earlier work (Masterson et al., 2008). Children who made proportionally more semantic errors also demonstrated better overall task performance, suggesting that such errors are associated with more advanced vocabulary knowledge. Most importantly, semantic errors foreshadowed later learning: children were more likely to correctly identify a word one year later if they had previously selected a semantically related distractor for that item.

Previous work has emphasized that word learning is not reducible to learning 1:1 mappings between words and objects. An adult-like understanding of word meaning emerges gradually over development and involves continuous refinement of a word’s conceptual space (Ameel, Malt, & Storms, 2008; Grosse, Streubel, Gunzenhauser, & Saalbach, 2021; Saji et al., 2011). The error patterns observed here suggest that even when children are not yet able to correctly identify a word’s referent, they possess partial semantic knowledge, in that hearing a word activates a structured semantic neighborhood. For example, when hearing the word “orca,” children may not correctly select an image of an orca but nevertheless show evidence of activation of a broader network of water-living animals. Such activation patterns might reflect the fact that words are not explicitly taught in isolation but learned in social-interactional contexts that contain a broad range of – linguistic and non-linguistic – information sources that provide clues to what a speaker means (Bohn & Casillas, 2026; Tomasello, 2008; Wojcik & Saffran, 2013). The current study further shows that this partial

semantic knowledge predicts later learning, suggesting that early-formed semantic structure scaffold subsequent word learning.

These longitudinal findings are particularly important because they show that semantic errors predict future learning over approximately one year of development on an individual level. Individual differences in language abilities are among the most reliable predictors of later life outcomes, including academic achievement, cognitive abilities, and psychological well-being (Bornstein, Hahn, Putnick, & Pearson, 2018; Marchman & Fernald, 2008; Morgan, Farkas, Hillemeier, Hammer, & Maczuga, 2015; Schoon, Parsons, Rush, & Law, 2010; Walker, Greenwood, Hart, & Carta, 1994). This raises two potential avenues for application. First, in addition to assessing vocabulary knowledge, vocabulary tasks that include semantic distractors may provide information about children’s learning potential through their error patterns, thereby offering a complementary measure of language development. Second, early language learning may be supported by strengthening children’s conceptual knowledge and by facilitating the development of structured semantic networks (Cervetti, Wright, & Hwang, 2016).

The present study is limited in that it investigated only a single language, namely German. To establish cross-linguistic generalizability, future work should test whether semantic errors also predict later learning in other languages. Tentative converging evidence comes from Kryzstofiak and colleagues (Kryzstofiak et al., 2025), who also found a cross-sectional association between the proportion of semantic errors and overall vocabulary performance in a Polish-speaking sample. More broadly, if semantic errors reflect properties of underlying conceptual organization rather than language-specific characteristics, similar predictive relationships should be observed across languages. A useful tool for investigating such cross-linguistic generalization are the Cross-Linguistic Lexical Tasks (Haman, Łuniewska, & Pomiechowska, 2015), which were explicitly designed to include semantic distractors and are available in a wide range of languages (Haman et al., 2017).

207 Taken together, these results suggest that systematic semantic errors not only index a
208 developing semantic network but also provide information about future learning (Reuter,
209 Borovsky, & Lew-Williams, 2019). More broadly, the present findings suggest that
210 conceptual knowledge is not only shaped by language learning but also actively scaffolds it,
211 such that the organization of semantic networks supports the acquisition of new words
212 (Borovsky, Ellis, Evans, & Elman, 2016).

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